

ALHASSAN JAWAD

ELECTRICAL ENGINEER • FOCUS: CYBER- & OT-SECURITY

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PROFESSIONAL SUMMARY

Electrical Engineer (B.Sc.) with 2+ years of academic experience as a Teaching Assistant at Uppsala University, specializing in technical evaluation and laboratory verification. Currently completing an M.Sc. in Electrical Engineering (expected Q1 2027) with coursework centered on OT-security and control of electrical systems.

Technical background bridges electrical infrastructure and control theory with programming and automation logic, applying signal processing and IoT frameworks to embedded architectures. Focus is directed toward engineering secure, resilient control systems for critical infrastructure.

TECHNICAL COMPETENCIES

Programming Languages: Python, Java, C/C++, JavaScript, HTML, CSS, MATLAB, LabVIEW
Engineering Tools & Frameworks: Git/GitHub, Angular, Altium Designer, PSpice, LTSpice, PostgreSQL
Engineering Skills: Automation, Control Systems, Signal Processing, Data Analysis
Languages: Swedish (Fluent), English (Fluent), Arabic (Native)

CERTIFICATIONS

- microI: Vetted Software Engineer (Electronic Engineering / Ngspice) [Issued 2026]
- Google Cloud Certified: Professional Google AI Engineer [Issued 2026]

RELEVANT EXPERIENCE

Software Engineer (Internship) – Portabel Health

Sep 2025 – Dec 2025

Programmed and debugged platform-level components in an Agile development pipeline using:

- Angular framework,
- Tailwind CSS,
- Node Package Manager (npm)

Executed 130+ technical tasks within an Agile development pipeline using Freedcamp, focusing on feature deployment and continuous testing across Dev and QA environments.

Managed backend data structures using PostgreSQL and translated UI/UX designs from Figma into functional code. To ensure high availability and prevent system downtime during updates, I wrote comprehensive script refactoring documentation for each new script, and created a standardized internal Git version control guideline.

Teaching Assistant – Uppsala University

Aug 2023 – Jul 2025

- Instructed and evaluated 50+ students per semester on core programming & electrical concepts.
- Was responsible for grading parts of each year's assignments and final exam.
- Collaborated with faculty and fellow peers to develop new course materials.

Co-founder & Chief Technology Officer (CTO) – NordicFlip / Furrytool

Mar 2023 – Jun 2024

Startups

Co-founded and engineered both startups from initial conceptual design to a functional small business platform, scaling the system to support approximately 100 global buyers prior to operations closing. Managed the full-stack technical lifecycle, executing all development tasks from initial system architecture design down to code-level implementation and platform deployment.



EDUCATION

Master of Science (M.Sc.) in Electrical Engineering

2025 – 2027

Uppsala University – Specialization: Control of Electrical Systems

Bachelor of Science (B.Sc.) in Electrical Engineering

2019 – 2022

Department of Electrical Engineering | Uppsala University

Degree issued in 2026

SELECTED PROJECTS

Master's Thesis: Automated Aeroponic Cultivation System

2025

Internal Collaboration with Uppsala University

- Programmed and deployed a multi-variable MIMO rule-based feedback control loop on an ESP32 microcontroller to regulate closed-loop temperature and humidity dynamics.
- Built and integrated a real-time IoT sensor telemetry network to establish a continuous data acquisition loop, thus verifying closed-loop feedback stability and firmware execution.
- Compiled technical documentation for system initialization, debugging, and multi-variable tuning routines.

Bachelor's Thesis: Smart Building Energy Management

2022

External Collaboration with Studenternas IP, Uppsala

- Wrote a custom optimization script in MATLAB to manage energy consumption and load distribution for a commercial property.
- Processed multi-variable datasets from deployed IoT sensors, isolating 71.2 kW of the facility's 185 kW total load as manageable power flexibility.
- Proved that 38.5% of the infrastructure's peak demand could be dynamically controlled and deferred without affecting facility operations.

Project: Miniature Elevator Prototype (Course: Project in Electronics)

2023

Academic Project | Uppsala University

- Designed a three-level miniature elevator, handling end-to-end HIL (Hardware-In-the-Loop) integration.
- Programmed the control logic and GUI (in LabVIEW), to manage real-time level selection and state transitions while regulating motor speed profiles, to achieve a smooth acceleration while minimizing physical oscillations.

Project: GPS Tracking Device (Course: PC-Board Construction with ECAD Tools)

2022

Academic Project | Uppsala University

- Designed a multi-layer custom PCB within Altium Designer.
- Integrated an ATmega328P microcontroller with a Quectel MC60 transceiver to process satellite GPS data and transmit coordinates over different networks.
- Implemented a hardware power-saving scheme by interfacing an MMA8653 accelerometer to provide system wake-up triggers based on physical motion.